G.O.S. sensor node rev 3.0

18650 cell + USB-C charging, nRF52840 module (Thread / BLE), SHT40 and TSL2591 over I2C.
LED drivers live on a separate board fed through J4. LoRa and GNSS are not fitted.

Sheets share nets through global labels and power symbols; the sheet symbols below carry no pins.
Pin map, power tree and datasheet sources: HARDWARE.md

Power

File: power.kicad_sch

MCU

File: MCU.kicad_sch

Sensors

File: sensors.kicad_sch

Connectors

File: connectors.kicad_sch

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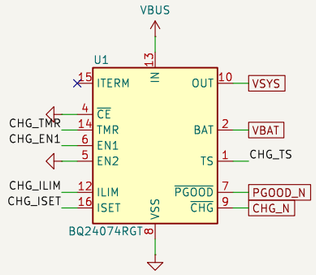
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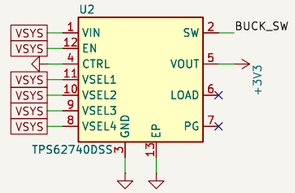
Sheet: /
File: gos_sensor_node.kicad_sch

Title: G.O.S. sensor node

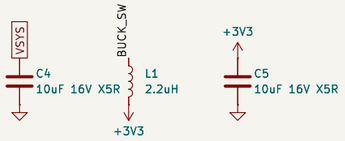
Size: A4	Date:	**Rev: 3.0**
KiCad E.D.A. 9.0.7		Id: 1/5

USB-C 5 V -> BQ24074 (charger + power path, VSYS = 4.4 V on USB, VBAT on battery) -> TPS62740 (3.3 V, all four VSEL high)
USB500 mode: EN2 = GND, EN1 high only while VBUS is present (no battery drain through the EN pull-down).
Charge 494 mA (ISET 1.8k), safety timer 9.1 h (TMR 68k), termination 10 % (ITERM open).
TH1 must sit against the cell holder. Without a thermistor fit a fixed 10k instead.

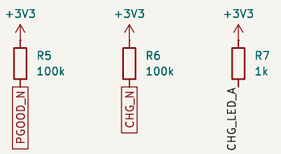


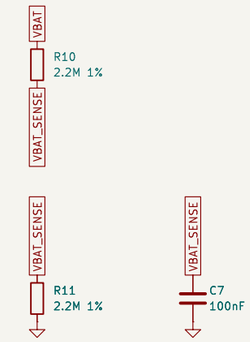






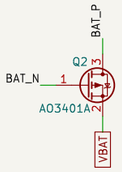


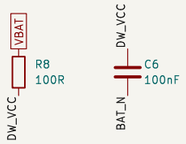


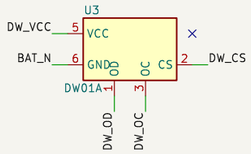


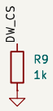
Cell protection. BAT_N is the cell negative, NOT system GND: the FS8205A sits between them.
DW01A trips: overcharge 4.30 V, overdischarge 2.40 V (release 3.0 V), overcurrent 150 mV across both FETs (about 2.5 A).
Q2 blocks a cell inserted backwards. After fitting a cell, plug USB in once to wake the DW01A.

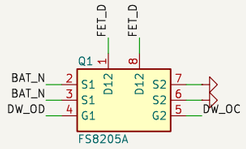










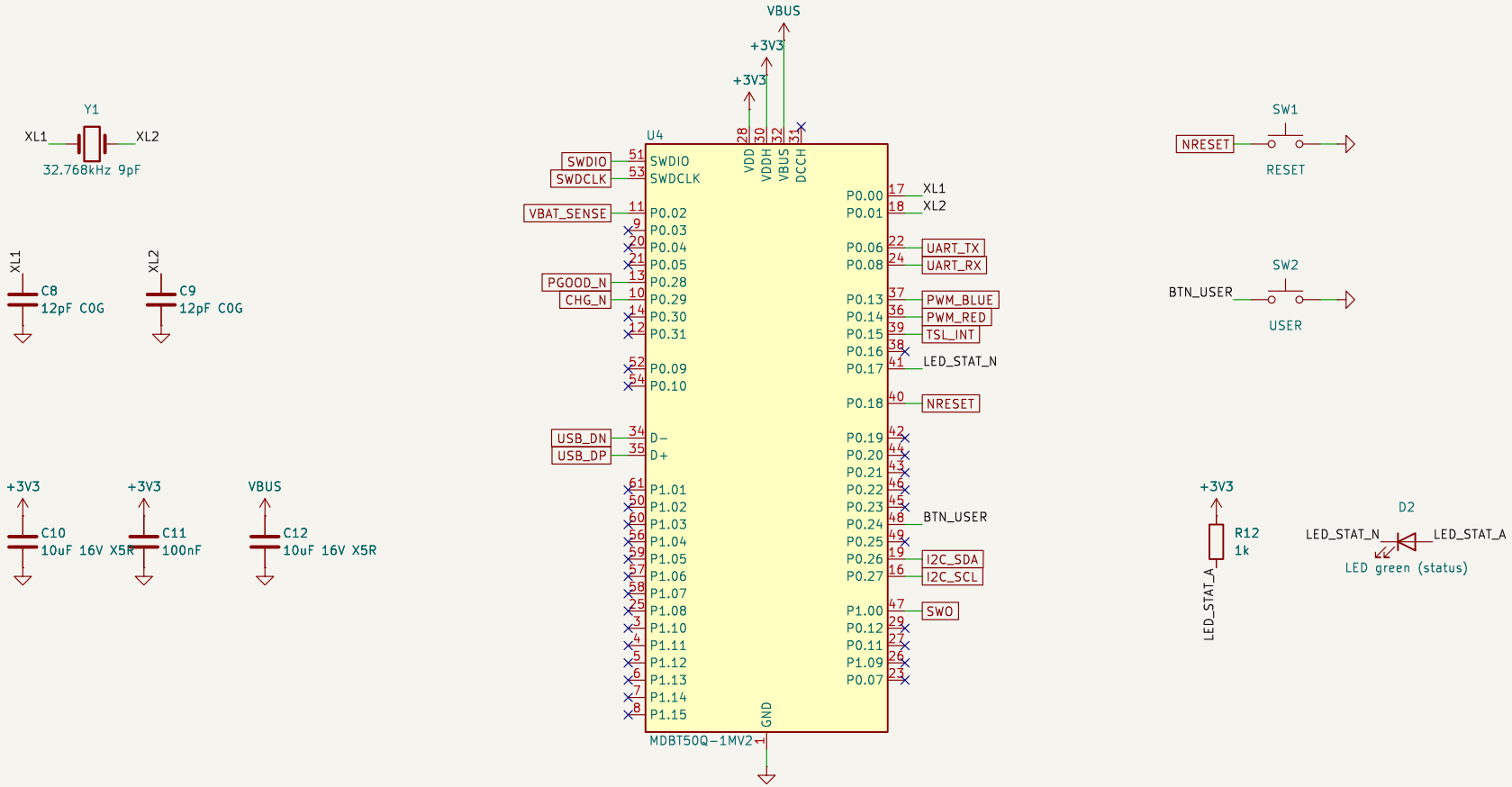


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Sheet: /Power/
File: power.kicad_sch

Title: Power: USB charger, 3.3 V buck, cell protection

Size: A3	Date:	**Rev: 3.0**
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VDD and VDDH tied to 3.3 V, DCCH open: Raytac reference circuit 8.3 (REG0 off, supply below 3.6 V).
Module pins 3–14 and 50–61 are low-frequency only (below 10 kHz): no SPI, I2C, UART or PWM there.
Y1: 32.768 kHz, 9 pF load, with 12 pF caps as in the Raytac reference.
Keep copper out of the module antenna area (Raytac spec section 2.3).

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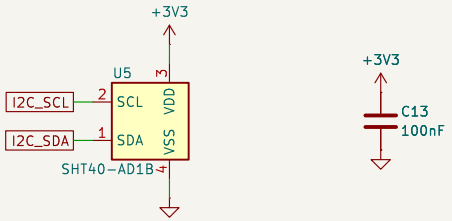
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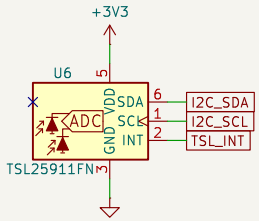
Sheet: /MCU/
File: MCU.kicad_sch

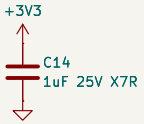
Title: MCU: Raytac MDBT50Q-1MV2 (nRF52840)

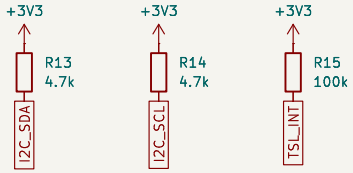
Size: A4	Date:	**Rev: 3.0**
KiCad E.D.A. 9.0.7		Id: 3/5

I2C addresses: SHT40 0x44, TSL2591 0x29. Place U5 away from the charger and buck (self-heating).
U6 needs a clear view of the light: no part taller than it nearby.









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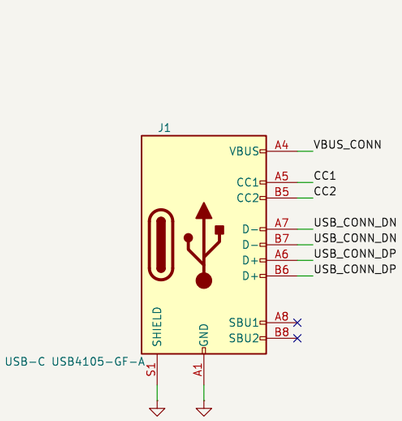
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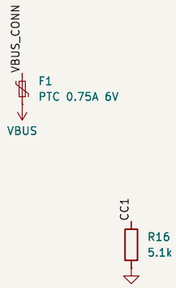
Sheet: /Sensors/
File: sensors.kicad_sch

Title: Sensors: SHT40, TSL2591

Size: A4	Date:	**Rev: 3.0**
KiCad E.D.A. 9.0.7		Id: 4/5

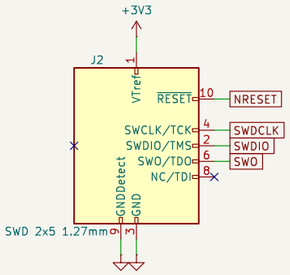
J1: USB 2.0 device. 5.1k on each CC pin asks the host for 5 V.
J4 feeds the separate LED driver board from VSYS (battery or USB). 100k pull-downs keep the drivers off during reset.
J2: standard 10-pin 1.27 mm Cortex debug header.

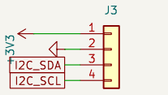


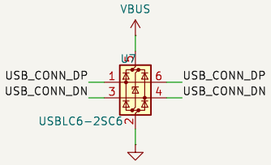




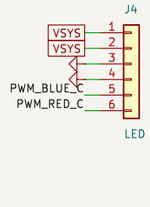




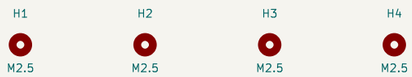












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Sheet: /Connectors/
File: connectors.kicad_sch

Title: Connectors: USB-C, SWD, I2C, LED board, test points

Size: A3	Date:	**Rev: 3.0**
KiCad E.D.A. 9.0.7		Id: 5/5